1.2.3 Twin Primes

A pair of prime numbers whose difference is 2, is called *twin primes*.

For example, (5, 7) is a twin prime pair as is (17,19).

Try to find a few more *twin prime pairs!*



If three successive prime numbers differ by 2, then the prime numbers form a prime triplet. The only prime triplet is (3, 5, 7).

1.3 Rules for Test of Divisibility of Numbers

Suppose that, you are asked to simplify a fraction say $\frac{126}{216}$. Since the numbers are relatively bigger, the task is not easy. Observe that, these numbers are not only divisible by 2 and 9 exactly but by other numbers too! How do we know that 2 and 9 are factors of 126 and 216? We are going to see ***divisibility tests*** in this section which are rules, that will improve your mental math skills for such determinations.

Divisibility tests, in common, are useful in the prime factorisation of a number. Also, it is fun to find whether any large number is exactly divisible by 2, 3, 4, 5, 6, 7, 8, 9, 10 or 11 (and more...) by simply observing, examining and performing basic operations with the digits of the given number and not by doing the actual division! Curious to know? Then, remember the following interesting rules and have fun...! As divisibility by 2, 3 and 5 gain importance in the prime factorisation of a number, we will define the rules for them first!

Divisibility by 2

A number is divisible by 2, if its ones place is any one of the even numbers 0, 2, 4, 6 or 8.

Examples:

1. 456368 is divisible by 2, since its ones place is even(8).
2. 1234567 is not divisible by 2, since its ones place is not even(7).

Divisibility by 3

Divisibility of a number by 3 is interesting! We can find that 96 is divisible by 3. Here, note that the sum of its digits 9+6 = 15 is also divisible by 3. Even 1+5 = 6 is also divisible by 3. This is called as *iterative* or *repeated* addition. So,

A number is divisible by 3 if the sum of its digits is divisible by 3.

Examples:

1. 654321 is divisible by 3.

   Here 6+5+4+3+2+1= 21 and 2+1=3 is divisible by 3.

   Hence, 654321 is divisible by 3.

6th_01_NUMBERS_T2_EM.indd 7 27-06-2020 04:41:30 PM